

correct production failures, and enables faster production ramps for continuous yield and cell efficiency improvement.

The system is highly modular in nature, so it is easy to scale and modify across the lifecycle, from research and development, to pilots, and then full production.

The system has an integrated database that contains all of the metrology measurements, quality measurements and process data necessary for one-stop engineering data analysis. And, an enterprise metadata manager for three-tier connectivity to any existing database, eliminating the need to duplicate any engineering and production data.

In January, Rudolph Technologies Inc. launched Discover Solar, a fabrication management software tool designed specifically to help photovoltaic manufacturers increase cell efficiency and reduce costs. The software is based on the company's data management software for the semiconductor industry, but incorporates a completely re-engineered database structure and analysis engine optimised for the high-volume photovoltaic production. It provides comprehensive analysis of process performance information, allowing manufacturers to act upon opportunities to improve the energy conversion efficiency of their products, monitor the health of a complete production line and quickly identify tool and sub-component problems, etc.

All you need is a single cell

The power produced by a single solar cell is too negligible to be put to conventional use.

At this year's Applied Power Electronics Conference and Exposition (APEC), Freescale Semiconductors, however, demonstrated an integrated mixed-signal chip that can convert the output from a single solar cell into a usable power source.

"Our power converter starts working at just 0.32V, allowing it to convert power from a variety of low-voltage



Freescale Semiconductor's Smartmos10 technology that can convert output from a single solar cell into a usable power source

sources," said Kevin Parmenter, applications engineering manager for the analogue, mixed-signal and power division of Freescale. "All sorts of green energy harvesting devices that had to be grouped together earlier to get their voltage up can now be used individually – from single solar cells to thermopiles, to bioelectrics."

Freescale claims that its power converter has 82 to 90 per cent efficiency. Thanks to its Smartmos mixed-signal process that allows digital, analogue and power devices to be housed on the same die. The key to use the 130nm Smartmos10 process for power converters was its ability to fabricate transistors that turned on at just 0.32V. For low-impedance connections to the circuit board, Freescale's power converters use a flip-chip on lead frame (FCOL) package. Freescale's power converter operates at 1MHz, permitting the use of a small inductor on the input despite the 1W power handling capability. This reduces the size of the converter.

New record in efficiency

In January 2009, a team from the Fraunhofer Institute for Solar Energy Systems announced that they have managed to squeeze an efficiency of 41 per cent out of photovoltaic (PV) cells, which they claimed to be a new world record.

The high efficiency was due to a specific technique used to grow the semiconductor material for the solar cells. They disclosed that the cells were

made of III-V-material on GaAs or Ge substrates. The technique of 'amorphous multiple cells' developed by the institute avoided the formation of impurity traps, which normally emerge in the transition zone of materials with different lattice constants.

This is a good example of how the control

over crystal defects within semiconductor materials can cause a technological breakthrough.

Another reason for the high efficiency was the scientists' choice of a material composition that created a spectral sensitivity distribution very similar to the spectral distribution of sunlight.

Sharper conversion by Sharp

Japan's New Energy and Industrial Technology Development Organisation (NEDO) launched a research initiative towards 'Innovative Solar Cells.' As part of this, Sharp achieved the world's highest solar cell conversion efficiency of 35.8 per cent in October this year. They did this using a triple-junction compound solar cell, which uses a stack of three photo-absorption layers made from compounds consisting of two or more elements such as indium and gallium – instead of the usual silicon-based solar cells.

In a compound solar cell, usually germanium (Ge) is used as the bottom layer due to its ease of manufacturing. However, a majority of the current generated by Ge is wasted, without being used effectively for electrical energy. Hence, the team replaced Ge with indium gallium arsenide (InGaAs), a material with high light utilisation efficiency. However, the process to make high-quality highly-crystalline InGaAs was difficult. Sharp has overcome this problem with its proprietary technology for forming layers. The amount of